$\det(xI - A) = (x-\lambda_1)^{r_{\lambda_1}} \cdots (x-\lambda_t)^{r_{\lambda_t}}$. $\{\lambda_1, \ldots, \lambda_t\}$ eigenvalues.

r_{λ_i} is the alg mult.

MA1522 Linear Algebra for Computing

Lesson 16: Eigenvalues and Eigenvectors

geo mult of A at λ_i is

$$\dim\left(\{v \in \mathbb{R}^n \mid Av = \lambda_i v\}\right) \longleftrightarrow r_{\lambda_i}$$

$\Updownarrow$

$(A - \lambda_i I)v = 0$

This is a fact, to prove this, need Cayley-Hamilton theorem.

$$\dim\left(\{v \in \mathbb{R}^n \mid (A-\lambda_i I)^k v = 0\}\right)$$

for some $k > 0$.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson16` (e.g. `A1234567Z_lesson16.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 16**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 6.1 Eigenvalues and Eigenvectors

Let A be a square matrix of order n.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \qquad xI - A = \begin{pmatrix} x-1 & -2 & -3 \\ -4 & x-5 & -6 \\ -7 & -8 & x-9 \end{pmatrix}.$$

Definition

A real number λ is an **eigenvalue** of A if there is a nonzero vector $\mathbf{v} \in \mathbb{R}^n$ such that $A\mathbf{v} = \lambda\mathbf{v}$.
Such a $\mathbf{v}$ is an **eigenvector** of A associated to λ.
The **characteristic polynomial** of A is the degree n polynomial $\det(xI - A)$.

$\lambda v = \lambda I v$

Theorem

λ is an eigenvalue of A if and only if λ is a root of the characteristic polynomial $\det(xI - A)$.

$\exists\, v \neq 0$ s.t. $Av = \lambda v \iff \exists\, v \neq 0$ s.t. $(\lambda I - A)v = 0 \iff \mathrm{Null}(\lambda I - A) \neq 0 \iff \lambda I - A$ is not invertible.

$\Downarrow$

$\iff \det(\lambda I - A) = 0$.

$\iff \lambda$ is a root of $\det(xI - A)$

Theorem

A is invertible if and only if 0 is not an eigenvalue of A.

$\det(A) \neq 0 \iff \det(0 \cdot I - A) \neq 0 \iff 0$ is not a root of $\det(xI - A)$

Section 6.1 Eigenspaces and Multiplicities

I will write $ch_A(x)$ for the char. poly of A.

Let λ be an eigenvalue of a square matrix A of order n.

Definition

The **algebraic multiplicity** r_λ of λ is the largest integer such that

$$\det(xI - A) = (x - \lambda)^{r_\lambda} p(x)$$

for some polynomial $p(x)$.

Definition

The **eigenspace** of A associated to λ is $(\lambda I - A)v = 0$

$$E_\lambda = \{ \mathbf{v} \in \mathbb{R}^n \mid A\mathbf{v} = \lambda\mathbf{v} \} = \text{Null}(\lambda I - A).$$

The **geometric multiplicity** of λ is $\dim(E_\lambda) = \text{nullity}(\lambda I - A)$.

Question 1

Remark: google "Jordan canonical form"

Let J be the Jordan canonical form with three blocks: a 2×2 block with eigenvalue 2, a 1×1 block with eigenvalue 2, and a 1×1 block with eigenvalue 3,

$$J = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

1. Find the characteristic polynomial of J.

$\det(xI - J) = \det \begin{pmatrix} x-2 & -1 & 0 & 0 \\ 0 & x-2 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 0 & 0 & 0 & x-3 \end{pmatrix}$

3 1

2 1

2. Find the geometric multiplicities of the eigenvalues 2 and 3.

If A is a n by n matrix, then $\det(xI - A) = (-1)^n \det(A - xI)$.

Question 1 — solution

1. $ch_J(x) = \det \begin{pmatrix} x-2 & -1 & 0 & 0 \\ 0 & x-2 & 0 & 0 \\ 0 & 0 & x-2 & 0 \\ 0 & 0 & 0 & x-3 \end{pmatrix} = (x-2)(x-2)(x-2)(x-3)$ prod of diagonal entries since this matrix is upper triangular

$= (x-2)^3 (x-3)^1$.

$\Rightarrow$ The eigenvalues of J are 2. 3.

with alg mult $r_2 = 3$ $r_3 = 1$

2. Compute the geometric mult at 2. 3. (geometric mult of λ = Nullity$(\lambda I - J)$)

(rank(A) + Nullity$(A) = n$)

$2I - J = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$ rank 2

$3I - J = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ rank 3.

$\Rightarrow$ Nullity$(2I - J) = 4 - 2 = \underline{2}$ $< 3 = r_2$ geo mult at $\lambda = 2$

Nullity$(3I - J) = 4 - 3 = \underline{1}$ geo mult at $\lambda = 3$.

Question 2

Let

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 3 & 0 & -3 \\ -1 & 1 & 4 \end{pmatrix}.$$

1. Find the characteristic polynomial of A, and factorise it.

2. Find the eigenvalues of A and their algebraic multiplicities. Is A invertible?

3. For each eigenvalue λ of A, find a basis for the eigenspace E_λ, and the geometric multiplicity of λ.

Question 2 — solution: part 1

$$\det(xI - A) = \det\begin{pmatrix} x-1 & -2 & -2 \\ -3 & x & 3 \\ 1 & -1 & x-4 \end{pmatrix} = (x-1)\det\begin{pmatrix} x & 3 \\ -1 & x-4 \end{pmatrix} - (-2)\det\begin{pmatrix} -3 & 3 \\ 1 & x-4 \end{pmatrix} + (-2)\det\begin{pmatrix} -3 & x \\ 1 & -1 \end{pmatrix}$$

$$= (x-1)\left(x(x-4)+3\right) + 2\left(-3(x-4)-3\right) - 2\left(3-x\right)$$

$$= (x-1)\left(x^2-4x+3\right) + (-6)(x-3) + 2(x-3)$$

$$= (x-1)(x-3)(x-1) + (-6)(x-3) + 2(x-3)$$

$$= (x-3)\left(x^2-2x+1-6+2\right)$$

$$= (x-3)\left(x^2-2x-3\right) = (x-3)^2(x+1).$$

by calculator

$ch_A(x) = x^3 - 5x^2 + 3x + 9$.

at least, calculator should tell you

$ch_A(x)$ has root $3, -1$.

$\Downarrow$

$(x-3)(x+1)$ divides $ch_A(x)$

$\Downarrow$

$x^3 - 5x^2 + 3x + 9 = \boxed{(x-3)(x+1)}(x-3)$.

Ans: The characteristic polynomial is $(x-3)^2(x+1)$

Question 2 — solution: part 2

2. Find the eigenvalues of A and their algebraic multiplicities. Is A invertible?

$ch_A(x) = (x-3)^2(x+1)^1$ → characteristic polynomial of A.

alg mult of $\lambda = 3$ is 2

$\lambda = -1$ is 1

A is invertible because 0 is NOT an eigenvalue.

Question 2 — solution: part 3, $\lambda = 3$

Compute the eigenspace $E_3 = \left\{ v \in \mathbb{R}^3 \;\middle|\; Av = 3v \right\} = \text{Null}(3I - A)$.

$$3I - A = \begin{pmatrix} 2 & -2 & -2 \\ -3 & 3 & 3 \\ 1 & -1 & -1 \end{pmatrix} \xrightarrow{\text{ref.}} \begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

set $x_2 = t$, $x_3 = s$

$t, s \in \mathbb{R}$

$$\begin{pmatrix} s+t \\ t \\ s \end{pmatrix} = t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \text{span}\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\} = E_3$$

geo mult $= \dim(E_3) = 2$.

Question 2 — solution: part 3, $\lambda = -1$

$$E_{-1} = \{ v \in \mathbb{R}^3 \mid Av = (-1)v \} = \mathrm{Null}\left((-1)I - A \right).$$

$$-I - A = \begin{pmatrix} -2 & -2 & -2 \\ -3 & -1 & 3 \\ 1 & -1 & -5 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\mathrm{Null}(-I - A) = \left\{ \begin{pmatrix} 2t \\ -3t \\ t \end{pmatrix} \;\middle|\; t \in \mathbb{R} \right\} = \mathrm{span}\left\{ \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \right\}$$

$$\begin{pmatrix} 2t \\ -3t \\ t \end{pmatrix} = t \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$$

basis for E_{-1}.

$\dim(E_{-1}) = 1$.

Homework

For each matrix below, find the characteristic polynomial and, for each eigenvalue λ, a basis for the eigenspace E_λ.

$$H_1 = \begin{pmatrix} 2 & 4 & 4 \\ 1 & -1 & -2 \\ -1 & -1 & 0 \end{pmatrix}, \qquad H_2 = \begin{pmatrix} 3 & 1 & -2 \\ 2 & 2 & 4 \\ -1 & 1 & 2 \end{pmatrix}, \qquad H_3 = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 5 & 0 \\ 1 & 2 & 3 \end{pmatrix},$$

$$H_4 = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 3 & 2 & 2 \\ 0 & 4 & -1 & 4 \\ 0 & 0 & -2 & 1 \end{pmatrix}, \qquad H_5 = \begin{pmatrix} 0 & 0 & 3 & 2 \\ 1 & 2 & 0 & -1 \\ 3 & 0 & -1 & -3 \\ -2 & 0 & 3 & 4 \end{pmatrix}.$$

All eigenvalues are integers.

Question 3

Let A be an $n \times n$ matrix. Suppose that λ is an eigenvalue of A.

k is integer.

1. Show that λ^k is an eigenvalue of A^k for $k \geq 1$. If A is invertible, show the same for $k \leq -1$.

2. We say a matrix A is **nilpotent** if $A^k = 0$ for some $k \geq 1$. Show that if A is nilpotent, then 0 is the only eigenvalue of A.

Question 3 — solution: part 1

If λ is an eigenvalue of A, then λ^k is an eigenvalue of A^k.

$\Updownarrow$

$\exists v \neq 0$ s.t. $\boxed{Av = \lambda v}$

- $k \geq 1$. We apply induction on k s.t. $A^k v = \lambda^k v$.

  $k=1$: automatic

  $k>1$: $A^k v = A^{k-1}(Av) = A^{k-1}(\lambda v) = \lambda(A^{k-1}v) = \lambda \cdot \lambda^{k-1} v = \lambda^k \cdot v$.

  ($Av = \lambda v$; induction hyp for $k-1$)

- $k \leq -1$ when A is invertible.

  $Av = \lambda v \implies A^{-1}Av = A^{-1}\lambda v$

  $v = \lambda A^{-1}v \implies \lambda^{-1}v = A^{-1}v$

Question 3 — solution: part 2

2. If A is nilpotent, then 0 is the only eigenvalue of A.

$A^k = 0$ for some k.

a) Show that 0 is an eigenvalue of A. $\Leftrightarrow$ A is not invertible.

$A^k = 0 \Rightarrow A^k$ is NOT invertible. $\Leftrightarrow$ $0 = \det(A^k) = \det(A)^k \Rightarrow \det(A) = 0$.

b) Show that all eigenvalues of A are 0.

Suppose λ is an eigenvalue of A.

I.e. $\exists v \neq 0$ s.t. $Av = \lambda v$.

By part 1, $\lambda^k v = \underbrace{A^k}_{0} \cdot v = 0 \Rightarrow \lambda^k = 0$ since $v \neq 0 \Rightarrow \lambda = 0$.

□

Challenge 1

Let a_0, a_1, a_2, a_3 be real numbers. Compute the characteristic polynomial of the following matrix:

$$C = \begin{pmatrix} 0 & 0 & 0 & -a_0 \\ 1 & 0 & 0 & -a_1 \\ 0 & 1 & 0 & -a_2 \\ 0 & 0 & 1 & -a_3 \end{pmatrix}.$$

Next lesson

Next lesson (**Fri 23 Oct**, Sections 6.2 & 6.3), I will focus on

1. diagonalization: when a square matrix A can be written as $A = PDP^{-1}$ with D diagonal, and how to find P and D;

2. **orthogonal diagonalization** of symmetric matrices.

Please watch the video clips beforehand.